

18.06 Problem Set 7

Due 11am Friday October 28.

Problem 1 (5+5+5 points)

(a) Consider the $m \times n$ matrix

$$A = \begin{pmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} \\ 1 & a_3 & a_3^2 & \cdots & a_3^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_m & a_m^2 & \cdots & a_m^{n-1} \end{pmatrix}$$

that we used in class for least-square fitting to polynomials, where a will be a vector of m equally spaced -points from 0 to 1. This is computed in Julia by:

```
a = range(0, 1, length=m)
A = a .^ (0:n-1)'
```

Construct this matrix A for $m = n = 100$, and plot its singular values `svdvals(A)` on a log scale with `using PyPlot, LinearAlgebra; semilogy(svdvals(A)); ylabel(L"\sigma_k"); title("...")` (always label your plots!).

You should observe that the singular values are rapidly decreasing - you could set all but about _____ of them (nearest multiple of 10) and still have nearly the same matrix A to ~ 16 decimal places - the matrix A is *nearly* rank _____ (nearest multiple of 10).

This means that many of the columns of A are nearly linearly **dependent**. Give a few words of explanation?

(b) Suppose that A is an $m \times m$ matrix with m *identical* singular values $\sigma_k = \sigma > 0$. It follows that A/σ must be a _____ (pick *one* of: singular, rank-deficient, projection, unitary, symmetric, uninteresting, diagonal, upper-triangular) matrix because _____.

(c) Plot the singular values of the matrix `A = rand(100,100)`, whose entries are random numbers distributed uniformly between 0 and 1. This matrix is *not* close to rank-deficient, because random columns are very unlikely to be close to linearly dependent, so you should find that none of the singular values are ridiculously small compared to the others. However, *one* singular value is much bigger than the others. Why? As a hint, plot the first 5 singular vectors v with `plot(svd(A).V[:,1:5])` - how is the singular vector for the biggest singular value different from the others? Maybe think about the best rank-1 approximation to this matrix implied by the SVD. (A few words of explanation suffice, no proofs required.)

Problem 2 (9+6 points)

(a) Apply Gram-Schmidt to the polynomials $1, x, x^2$ to find an orthonormal basis q_1, q_2, q_3 of polynomials under the inner ("dot") product (*different* from the one used in class):

$$f \cdot g = \int_0^\infty f(x)g(x)e^{-x}dx$$

(Unlike the Legendre polynomials in class, normalize your polynomials q_k to have $\|q_k\| = \sqrt{q_k \cdot q_k} = 1$ under this inner product, i.e. to be really *orthonormal*.)

- The following integral will be useful: $\int_0^\infty x^n e^{-x} dx = n!$ (n factorial) for any integer $n \geq 0$.

(b) Consider the function $f(x) = \begin{cases} x & x < 1 \\ 0 & x \geq 1 \end{cases}$. Find the *degree-1 polynomial*

$p(x) = \alpha x + \beta$ that is the "best fit" to $f(x)$ in the sense of minimizing

$$\|f - \alpha x - \beta\|^2 = \int_0^\infty [f(x) - p(x)]^2 e^{-x} dx$$

In particular, find $p(x)$ by performing the orthogonal projection (with this dot product) of $f(x)$ onto

Problem 3 (5+5+5 points)

In a common variant of least-squares fitting called "ridge" or "Tikhonov" regression to get more robust solutions to noisy fitting problems, one minimizes:

$$f(x) = \|b - Ax\|^2 + \alpha \|x\|^2$$

instead of just $\|b - Ax\|^2$, for some "penalty" parameter $\alpha \geq 0$. As α gets bigger and bigger, this favors *smaller* solutions x . Here, A is $m \times n$.

(a) Write this *same* $f(x)$ as $f(x) = \|c - Bx\|^2$ for some matrix B and some vector c defined in terms of A, α, b . Hint: $\|x\|^2 + \|y\|^2$ equals the length² of a single vector _____ made by stacking x and y .

(b) Since part (a) rewrote $f(x)$ as an ordinary-looking least-squares problem, use the normal equations for B and c to write down an equation $(n \times n \text{ matrix})\hat{x} = (\text{right-hand side})$ for the minimizer $\hat{x}$ in terms of A, α, b .

(c) For $\alpha > 0$, show that (b) has a *unique* solution even if A is *not* full column rank (i.e. $A^T A$ is singular). In particular, why does your matrix on the left-hand-side have a nullspace of $\{0\}$?

A really common strategy in mathematics is to try to rewrite new problems so that they look like old problems, letting us re-use the old solutions!

Problem 4 (2+2+2+2 points)

Suppose we have an $m \times 3$ matrix $A = (a_1 \ a_2 \ a_3)$ (of rank 3), and we do Gram-Schmidt to obtain a matrix $Q = (q_1 \ q_2 \ q_3)$, or equivalently the QR factorization $A = QR$. In the following questions, **simplify your answers as much as possible** (in terms of the given matrices and/or vectors).

(a) What is $Q^T A$?

(b) If P_Q is the projection matrix onto $C(Q)$, what is $P_Q A$?

(c) If P_A is the projection matrix onto $C(A)$, what is $P_A Q$?

(d) What is $P_Q B$ where $B = (2a_1 \ 3q_2 \ 4a_3 - q_1)$?

Problem 5 (5+5 points)

The 3×3 matrix $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ (assumed full rank / invertible) has the QR factorization

$$A = Q \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ & r_{22} & r_{23} \\ & & r_{33} \end{pmatrix}$$

(a) Give a formula for r_{11} in terms of $a, b, c, d, e, f, g, h, i$ (but not any of the elements of Q).

(b) Solve for the vector x in the equation

$$Q^T x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

in terms of $a, b, c, d, e, f, g, h, i$.